Name. I. Abhinay H.No:2303A51811 Batch:26

| SCHOOL OF COMPUTER SCIENCE AND ARTIFICIAL INTELLIGENCE | | DEPARTMENT OF COMPUTER SCIENCE ENGINEERING | |
|---|---|---|---|
| **Program Name:**B. Tech | | **Assignment Type: Lab** | **Academic Year:**2025-2026 |
| **Course Coordinator Name** | Dr. Rishabh Mittal | | |
| **Instructor(s)Name** | Mr. S Naresh Kumar<br>Ms. B. Swathi<br>Dr. Sasanko Shekhar Gantayat<br>Mr. Md Sallauddin<br>Dr. Mathivanan<br>Mr. Y Srikanth<br>Ms. N Shilpa<br>Dr. Rishabh Mittal (Coordinator)<br>Dr. R. Prashant Kumar<br>Mr. Ankushavali MD<br>Mr. B Viswanath<br>Ms. Sujitha Reddy<br>Ms. A. Anitha<br>Ms. M.Madhuri<br>Ms. Katherashala Swetha<br>Ms. Velpula sumalatha<br>Mr. Bingi Raju<br>Mr. G. Kranthi | | |
| **Course Code** | 23CS002PC304 | **Course Title** | AI Assisted Coding |
| **Year/Sem** | III/I | **Regulation** | R23 |
| **Date and Day of Assignment** | Week 7 - Thursday | **Time(s)** | 23CSBTB01 To 23CSBTB52 |
| **Duration** | 2 Hours | **Applicable to Batches** | All Batches |
| **AssignmentNumber:14.4** (Present assignment number)/**24**(Total number of assignments) | | | |

| Q.No. | Question | *ExpectedTime to complete* |
|---|---|---|
| 1 | **Lab 14: Web Frontend Development – AI-Assisted HTML/CSS/JS Generation** | Week 7 |

**Lab Objectives**

- Design **functional, visually appealing** web applications using HTML, CSS, and JavaScript with AI assistance.
- Apply **responsive, accessible, and interactive design principles**.
- Create **practical UI components** for real-world web applications.
- Use AI to optimize **layout, UX, and performance**.

**Lab Outcomes**

- Applying AI assistance to frontend development tasks
- Understanding how HTML, CSS, and JavaScript work together
- Improving responsiveness, interactivity, and usability
- Evaluating and refining AI-generated frontend code

# Task 1: Responsive Homepage Layout for a Startup Website

## Scenario

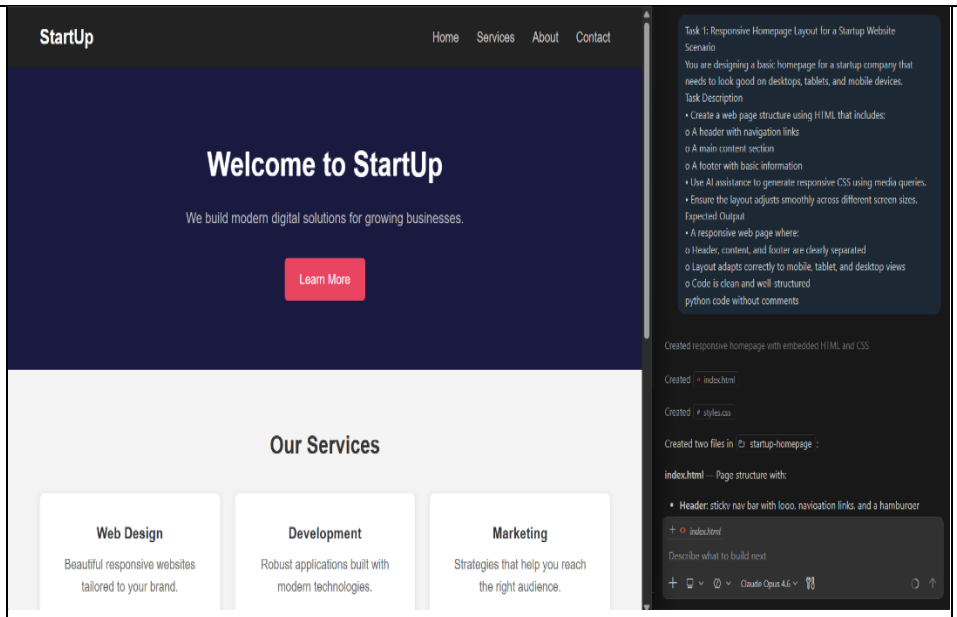
You are designing a **basic homepage for a startup company** that needs to look good on desktops, tablets, and mobile devices.

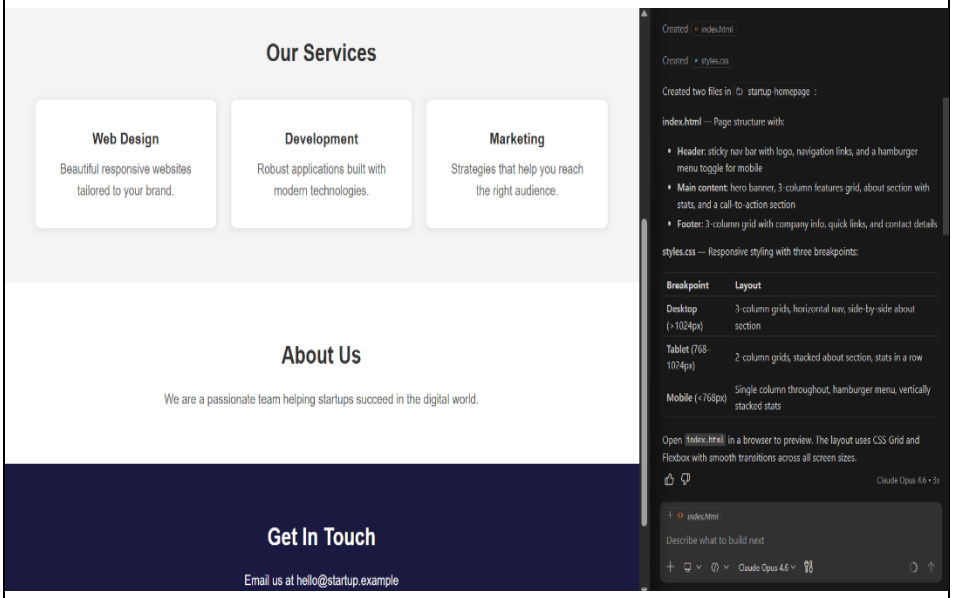
## Task Description

- Create a web page structure using HTML that includes:
  - A header with navigation links
  - A main content section
  - A footer with basic information
- Use AI assistance to generate responsive CSS using media queries.
- Ensure the layout adjusts smoothly across different screen sizes.

## Expected Output

- A responsive web page where:
  - Header, content, and footer are clearly separated
  - Layout adapts correctly to mobile, tablet, and desktop views
  - Code is clean and well-structured




# Task 2: Interactive Call-to-Action Button

## Scenario

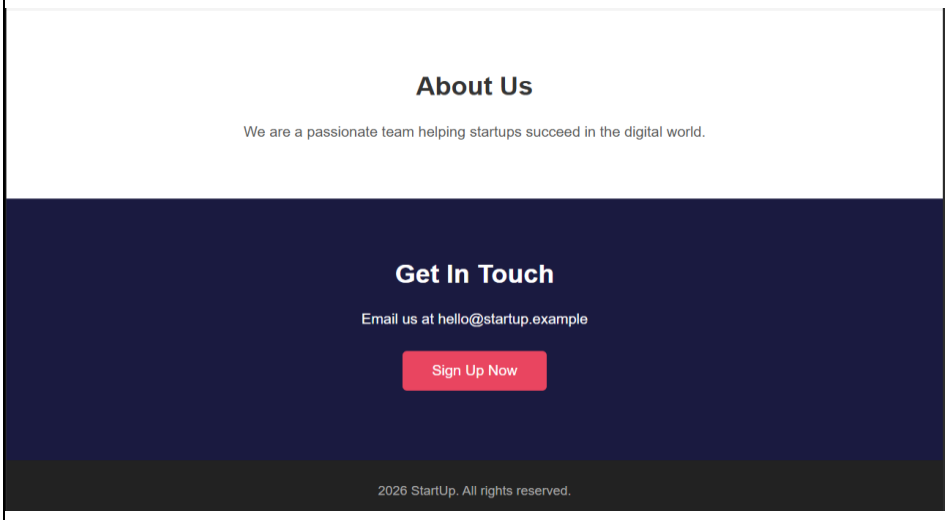
A marketing page requires a **call-to-action button** that responds when users interact with it.

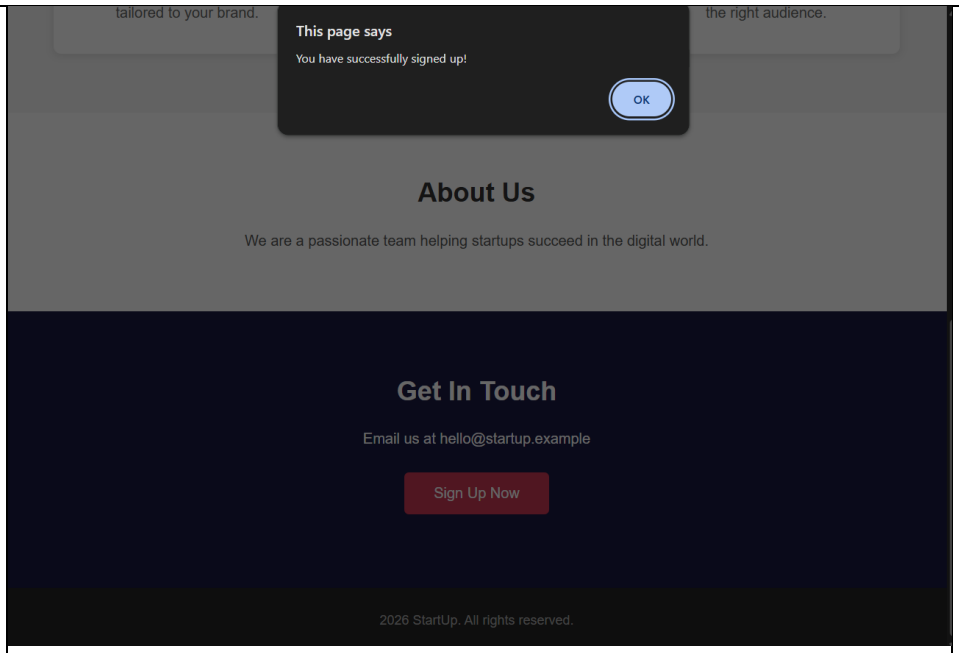
## Task Description

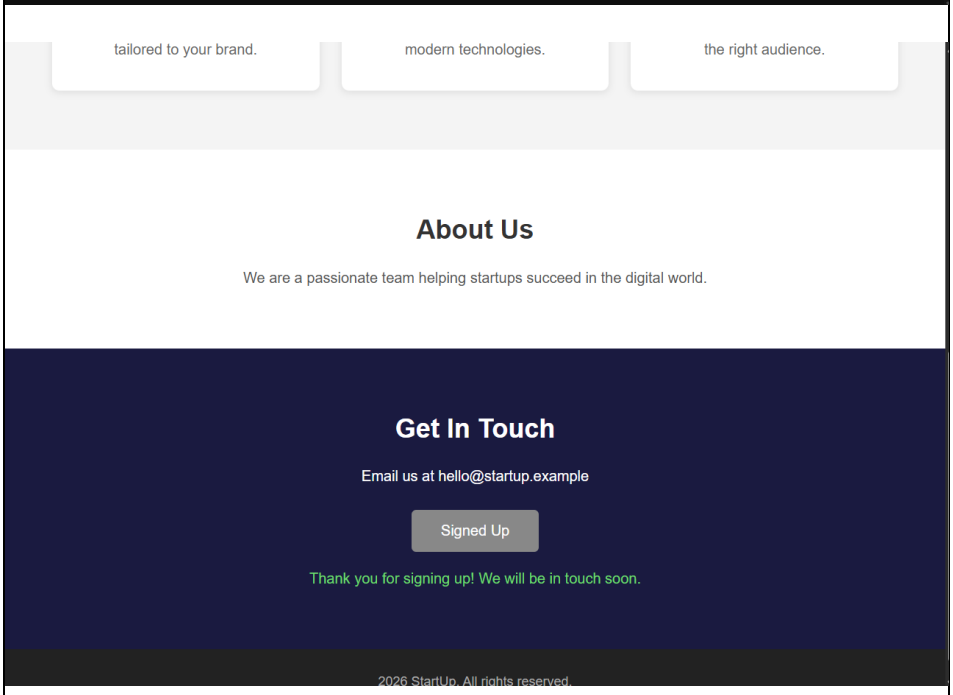
- Add a button to a web page using HTML.
- Use AI to generate JavaScript that responds to the button click by:
  - Displaying a message or alert
- Ask AI to add meaningful comments explaining the JavaScript logic.

## Expected Output

- A web page where:
  - Clicking the button triggers a visible response (alert or message)
  - JavaScript code is readable and commented
  - No page reload occurs






# Task 3: Contact Form with Client-Side Validation

## Scenario

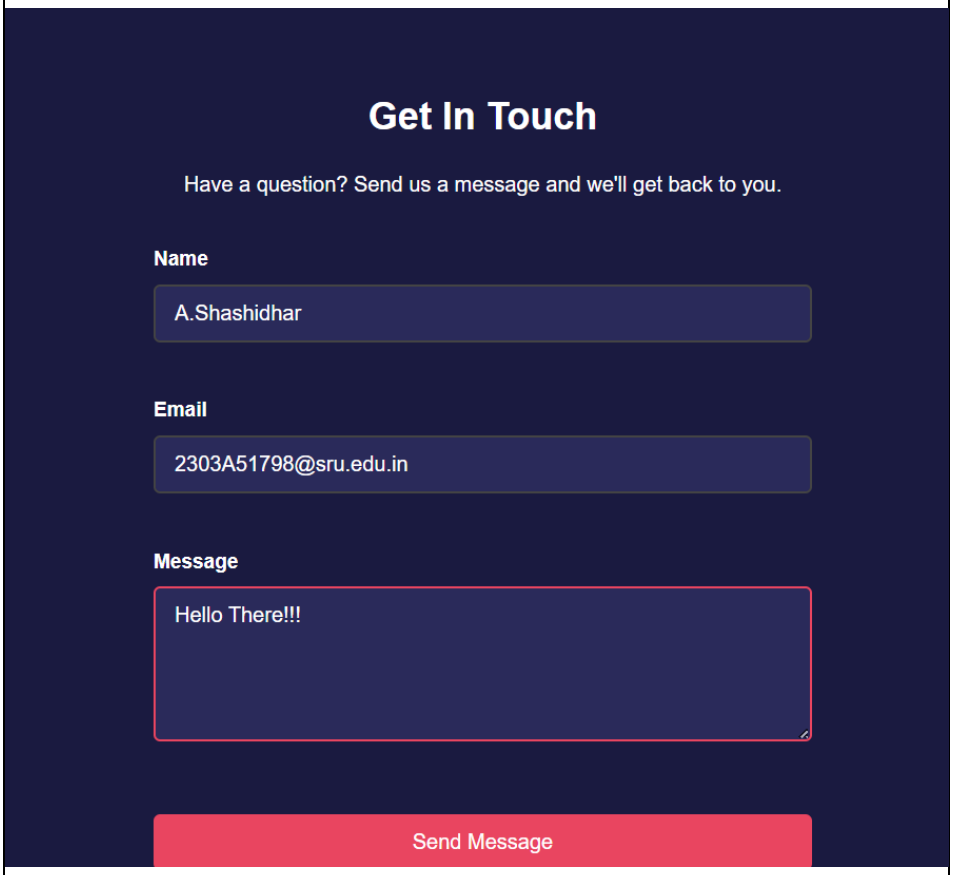
You are developing a **contact form for a company website** to collect user queries safely and correctly.

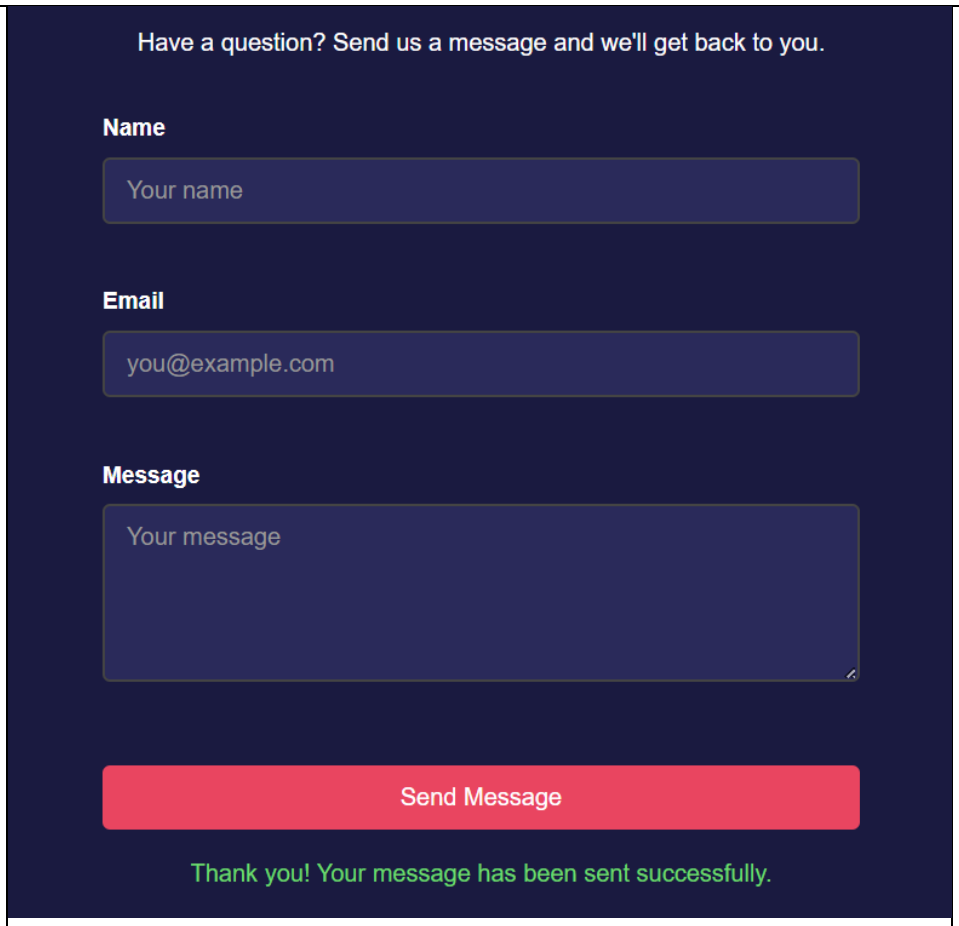
## Task Description

- Design a form with fields for:
  - Name
  - Email
  - Message
- Use AI to generate JavaScript validation logic that:
  - Prevents empty submissions
  - Checks for a valid email format
- Display clear error messages for invalid input.

## Expected Output

- A functional form where:
  - Invalid input shows inline error messages
  - Valid input displays a success message
  - Validation happens on the client side using JavaScript






# Task 4: Dynamic List Management for a Product Page

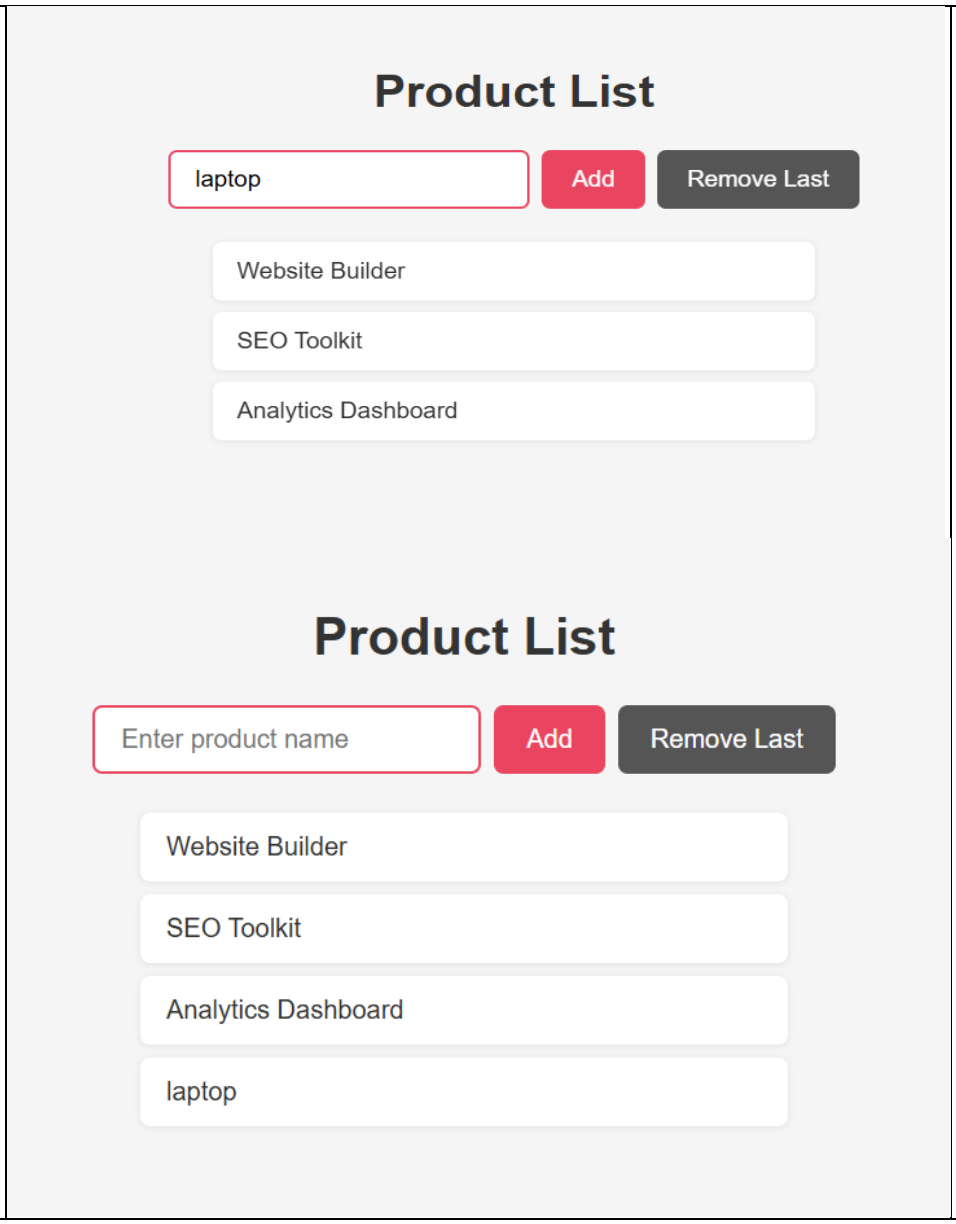
## Scenario

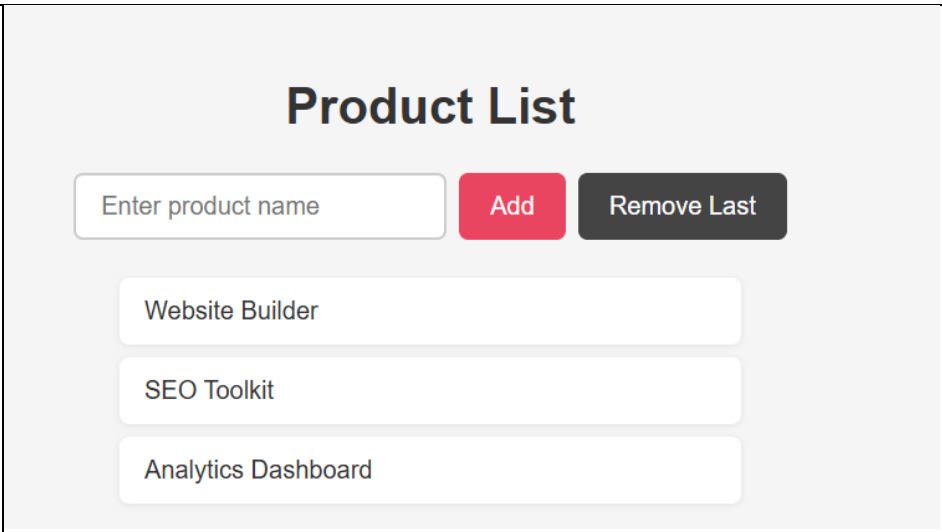
A product listing page needs to **add or remove items dynamically** without refreshing the page.

## Task Description

- Create a simple HTML list to display items.
- Use AI to generate JavaScript that:
  - Adds a new item when an “Add” button is clicked
  - Removes the last item when a “Remove” button is clicked
- Ensure the DOM updates dynamically.

| | | |
|---|---|---|
| | **Expected Output**<br><br>• A web page where:<br>  ○ Items can be added and removed dynamically<br>  ○ Changes appear instantly without page reload<br>  ○ JavaScript logic is clear and maintainable | |


Product List
laptop
Add
Remove Last
Website Builder
SEO Toolkit
Analytics Dashboard
Product List
Enter product name
Add
Remove Last
Website Builder
SEO Toolkit
Analytics Dashboard
laptop




# Task 5: Modal Popup for User Notifications

## Scenario

You are building a **user notification popup** for announcements or alerts on a website.

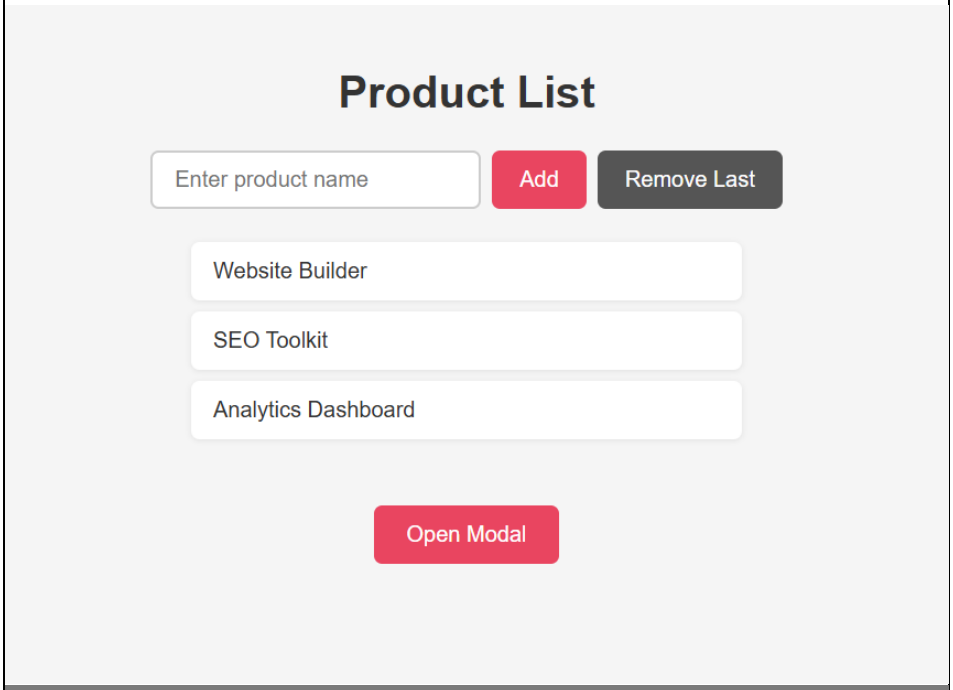
## Task Description

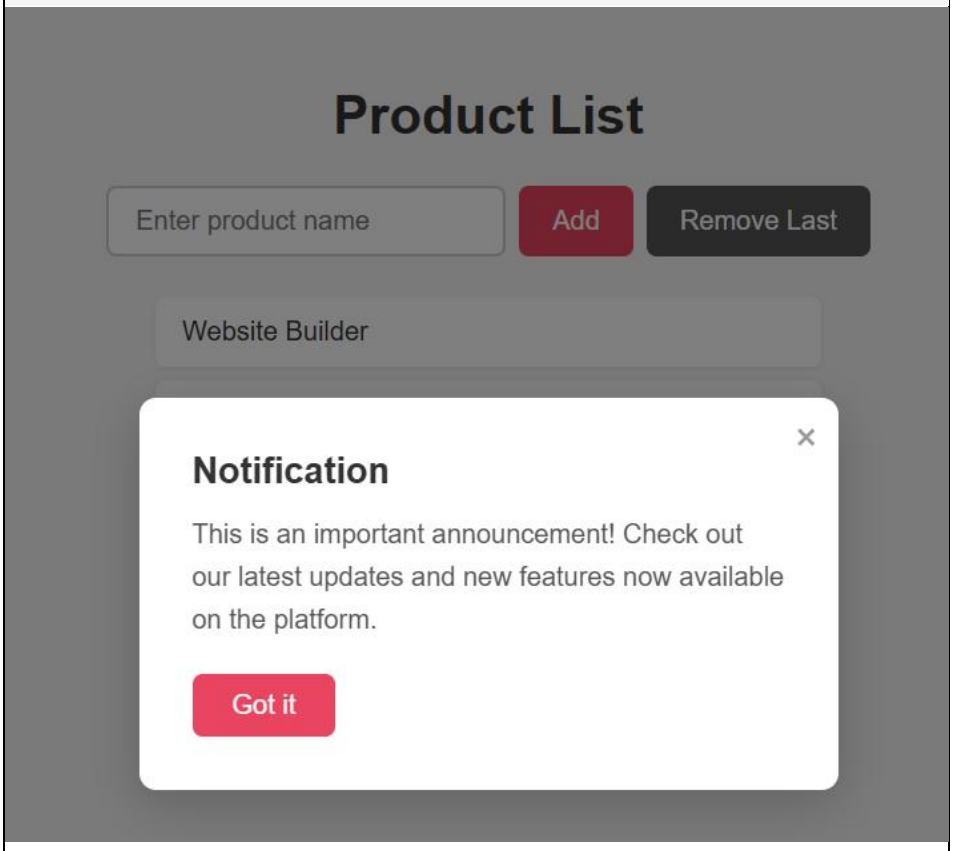
- Use AI to help generate:
  - HTML structure for a modal popup
  - CSS styling with a semi-transparent background overlay
  - JavaScript to open and close the modal
- Include multiple ways to close the modal (button or overlay click).

## Expected Output

- A web page where:
  - Clicking “Open Modal” displays the popup
  - Popup overlays the page content clearly

- Modal closes smoothly when instructed




**Note: Report should be submitted a word document for all tasks in a**

| | | |
|---|---|---|
| | **single document with prompts, comments & code explanation, and output and if required, screenshots** | |